

Parallels in the regulatory landscape of dimorphic female and male genital structures in *Drosophila melanogaster*

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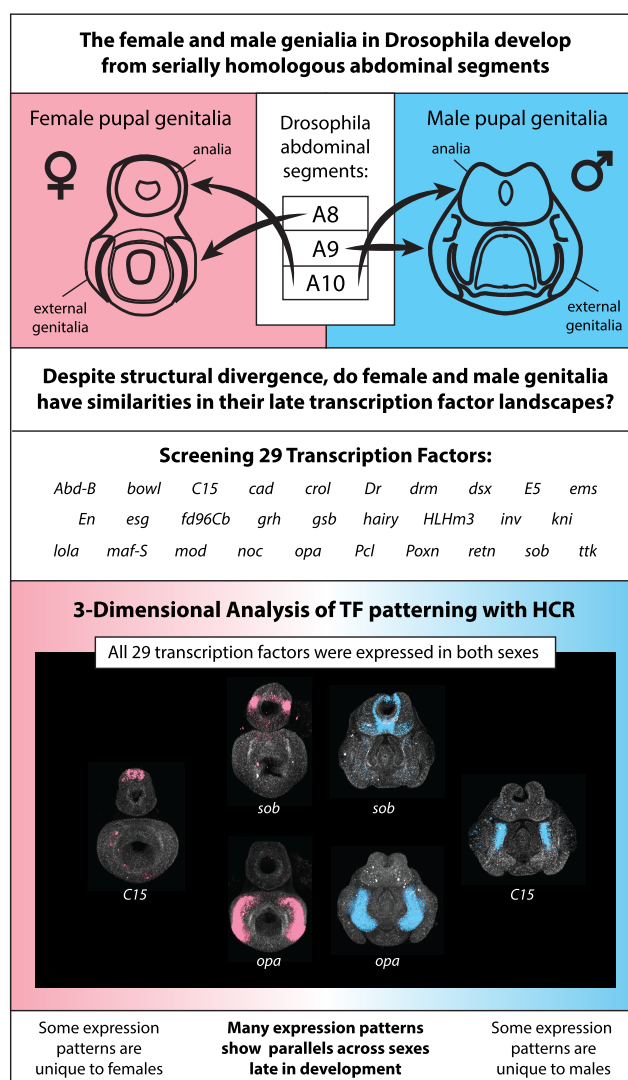
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Understanding how morphological structures evolve via changes to their development is an ongoing pursuit in biology. Comparative approaches examine changes in the expression or function of key developmental molecules within homologous structures and correlate these changes with structural divergence across species, populations, the sexes, or even between different body parts within individuals. The female and male genitalia of *Drosophila* offer an excellent opportunity to investigate homology and trait evolution, as fruit fly genital structures are developmentally tractable and evolve rapidly. While previous work has characterized gene regulatory networks operating in the development and evolution of male genital structures in *Drosophila*, female pupal genitalia are comparatively understudied. Here, we traced the development of female pupal genitalia to determine when and how individual structures form. We then measured the expression patterns of 29 transcription factors in both female and male genital structures at high resolution using hybridization chain reaction and confocal microscopy. We found that these transcription factors are expressed in both sexes, and some serve as marker genes for distinct genital structures in females. Our results suggest that the same transcription factors may control developmental processes in female and male genitalia, and this data enables future studies that interrogate how developmental gene regulatory networks specialize and evolve in both sexes.

Graphical Abstract



Keywords: *Drosophila*; terminalia; development; genitalia; evolution

Introduction

Demystifying the developmental processes that produce organismal structures and applying that knowledge to understand how those structures evolve is an ongoing pursuit in biology. Comparative approaches are a cornerstone of this field: changes to key molecules (eg transcription factors, signaling molecules, or cellular effectors) within homologous structures can be correlated with structural divergence across species, populations, the sexes, or even between different body parts within individuals (Smith et al. 2018). This approach works well for many intuitively homologous features such as mammalian digits (Cooper et al. 2014; Lopez-Rios et al. 2014) or serially homologous segments such as vertebrae (Burke et al. 1995) and crustacean limbs (Averof and Patel 1997). However, homology relationships are not always straightforward to identify. Sexually dimorphic traits, deep homology of long-diverged structures across species, or drastically diversified serial homologs can be challenging to recognize because evolutionary changes can mask underlying

similarities (DiFrisco and Jaeger 2021). Furthermore, the emergence of evolutionary novelties often involves network co-option: the redeployment of a pre-existing gene regulatory network in a new developmental context (McQueen and Rebeiz 2020; DiFrisco et al. 2023). This mechanism can produce similar regulatory networks that control the development of nonhomologous structures (McQueen and Rebeiz 2020; DiFrisco and Jaeger 2021).

Whenever a homology relationship exists between structures that must develop from, and be inherited within, the context of the same genome (ie sexually dimorphic traits, serial homology, network co-option), pleiotropy may influence how these structures evolve (Angelini et al. 2012). Thus, identifying even highly cryptic cases of homology is critical for understanding the evolution of forms. For these types of features, trait-specific molecular markers have been used to assign homology and facilitate evolutionary analysis. In particular, structures and cell types can be defined using expression patterns for genes that serve key functions in development, and shifts in the presence, timing, or spatial

patterns of these molecular markers can be compared across groups (Moczek and Rose 2009; Prud'homme et al. 2011; Hu et al. 2019; Fisher et al. 2020). However, the use of molecular methods to identify cryptic homology relationships and demarcate organismal features is still emerging in the field of evolutionary biology (Rice et al. 2023).

The female and male genitalia of *Drosophila* offer an excellent opportunity to gain new insights into the study of homology and trait evolution using comparative developmental approaches. Due to their rapid diversification, animal genitalia have long been used as models to address important questions in both evolutionary and developmental biology (Eberhard 1985; Hosken and Stockley 2004; Simmons 2014; Sloan and Simmons 2019). In *Drosophila*, the genitalia of both females and males are structurally diverse and rapidly evolving (Yassin and Orgogozo 2013; Kamimura 2016; Muto et al. 2018; Green et al. 2019; Onuma et al. 2022; Rice et al. 2023; Urum et al. 2024). In several cases, the rapid evolution of *Drosophila* genital structures has been found to involve developmental network co-option (Glassford et al. 2015; Molina-Gil et al. 2023; Rice et al. 2024). The preponderance of historical research in Drosophilids and other species has focused on the male genitalia (Méndez and Córdoba-Aguilar 2004; Ah-King et al. 2014; Orbach 2022). However, because female and male genitalia develop from serially homologous segments (Keisman et al. 2001), closing this sex-bias through further investigation of both sexes opens exciting opportunities to study multiple interconnected levels of homology relationships (Yassin and Orgogozo 2013; Ah-King et al. 2014; Brennan and Prum 2015; Brennan 2016).

In *Drosophila melanogaster*, previous studies have carefully examined adult genital morphology of males (Rice et al. 2019; Urum et al. 2024) and females (McQueen et al. 2022). Still, challenges arise for comparative approaches involving female genital traits in *Drosophila*, as boundaries between some key genital features are less distinct compared to the male, and not all of the features that vary from species to species are rigid or external (Yassin and Orgogozo 2013; McQueen et al. 2022). Nascent adult structures for females are especially difficult to distinguish at earlier developmental stages (Green et al. 2019). In addition, early studies of *Drosophila* genital development were performed on imaginal discs in third instar larvae, but determining how these tissues are transformed into the adult genital structures in both sexes remains challenging (Chatterjee et al. 2011).

Modern molecular methods can overcome the many obstacles inherent in investigating the evolution of complex genital structures. The application of in situ hybridization, immunohistochemistry, and other genetic techniques can aid in structural demarcation, even in early stages of genital development when boundaries are less distinct (Green et al. 2019; Smith et al. 2020). In previous work, we characterized the transcription factor landscape that patterns developing male genitalia in *D. melanogaster* (Vincent et al. 2019). We identified several transcription factors that serve as markers for genital structures with clear boundaries, and also found evidence that some structures may contain multiple cell types with unknown functions. This work demonstrated that the developing male terminalia are patterned by an intricate network of transcription factors and showed that some of these genes are restricted to a few structures or cell types. However, colorimetric in situ hybridization does not allow for precise visualization of expression patterns for internal structures, nor is it capable of capturing the relative spatial arrangement of multiple transcription factors within the same sample. Addressing these limitations will advance our ability to examine homology and structural evolution in this system using molecular methods.

In this study, we used hybridization chain reaction (HCR) and confocal microscopy to visualize the gene expression patterns of transcription factors in the genitalia of both sexes, with a particular focus on the female (Dirks and Pierce 2004; Choi et al. 2018). Our goal is to broaden our understanding of complex homology relationships and pave the way for future investigations in this powerful model system. Compared to colorimetric in situ hybridization, HCR enables greater sensitivity and resolution in 3 dimensions. In addition, HCR can identify putative regulatory connections between genes because it allows for the simultaneous detection of multiple gene expression patterns in the same sample (Schwarzkopf et al. 2021; Duckhorn et al. 2022). Using this technique, we found that many transcription factors are expressed in both female and male genital tissues, and that multiple structures in the female can be subdivided into domains with distinct genetic profiles. Many of the expression patterns we observe here corroborate tissue domain delineations that were previously proposed based on a taxonomically-informed analysis of the adult morphology (McQueen et al. 2022). Our experiments illuminate the cryptic complexity of the developing female genitalia and identify molecular markers that may inform future studies on the function and evolution of genital structures in both sexes.

Materials and methods

Drosophila strains

We obtained our lines from the Bloomington *Drosophila* Stock Center: yellow white, y^1w^1 , (Bloomington stock number #1495) and armadillo-green fluorescent protein (GFP), arm-GFP, (Bloomington stock number #8556). These stocks can be ordered from Bloomington.

Sample collection and dissection

Female and male *D. melanogaster* pre-pupa were collected at room temperature and incubated in a petri dish with moisture for 48 h prior to dissection. The posterior tip of the pupa was then separated, and the terminalia flushed out using a micropipette. The samples were then collected in phosphate-buffered saline (PBS) and then fixed in PBT and 4% paraformaldehyde (PFA). Samples were finally washed 3 times in PBT and stored at 4 °C for less than 12 h before being processed for the HCR reaction. Arm-GFP samples were mounted and imaged after fixation.

Hybridization chain reaction (HCR)

Samples were taken from storage and blocked by incubating in antibody buffer for 2 h at room temperature. They were then incubated with rat anti-E-cadherin, 1:100 in PBT (DSHB Cat# DCAD2, RRID:AB_528120) overnight at 4 °C. The samples were then washed 3 times in PBT, then washed 4× for 15 min in PBT and incubated with Molecular Instruments antibodies anti-rat B4 secondary antibody at 1 µg/mL for 2 h at room temperature. Samples were then washed with PBT 3× and post-fixed for 10 min in 4% PFA in PBT before being washed 3 more times in PBT. It was then washed once in 5× sodium chloride–sodium citrate with 0.1% Tween (5× SSCT). Probe solutions made with 8 µL HCR initiator probe set per 250 µL of incubation volume. RNA isoforms for each target gene were identified using FlyBase (release FB2023_05; Öztürk-Çolak et al. 2024). All HCR probes were designed and synthesized by Molecular Instruments Inc. Transcript accession numbers and sequences that were sent to Molecular Instruments, Inc. for HCR probe design of each gene can be found in [Supplementary Table 2](#). Samples were prehybridized in probe hybridization buffer for 30 min at 37 °C before being

incubated in 250 μ L of the probe solution for 48 h at the final hybridization step. Samples are then washed 4 times with probe wash buffer and 3 times with 5 $\times$ SSC. B1 488, B3 647, and B4 546 amplifiers were used. h1 and h2 amplifier hairpins are snap-cooled by heating 5 μ L of hairpin stock per sample to 95 $^{\circ}$ C for 90 s then cooling for 30 min in the dark to room temperature. Samples are then preamplified for 30 min in an amplification buffer and then amplified overnight in the dark in an amplification solution made of snap-cooled hairpins and amplification buffer. Finally, samples are washed 3 times in PBT and once in a 50:50 PBT glycerol solution. Samples are then moved to glycerol and mounted on microscope slides coated with poly-L-lysine.

Imaging and image processing

Samples were imaged at 20 $\times$ on a Leica TCS SP8 confocal microscope. For HCR, the expression patterns of 2 genes and ECAD, which gave only weak expression, were taken for all samples. As the imaged structures are 3-dimensional in nature, we used the MorphoGraphX program (Barbier de Reuille et al. 2015) to render and manipulate images in 3 dimensions. This allowed us to rotate the samples to better present the most informative perspectives of the various phallic structures. Raw image files are available on request.

Results

Developmental time course of the female genitalia

The female genitalia of *D. melanogaster* develop primarily from the eighth abdominal segment (Epper and Nöthiger 1982). The ninth abdominal segment is reduced in females and is the progenitor of only a few internal genital structures (Keisman et al. 2001; Sánchez et al. 2001). Notably, the opposite is true in the case of the male genitalia—the eighth segment is reduced in males, and most genital structures form from the ninth segment (Schüpbach et al. 1978; Epper and Nöthiger 1982). The analia form from the tenth abdominal segment in both sexes (Nöthiger et al. 1977). Here, we examined the morphology of female genitalia during 44 to 56 h after puparium formation (APF) pupal development, which includes timepoints during which species-specific differences have been previously traced (Green et al. 2019). These timepoints follow the evagination of the apical terminalia from their initial internal orientation (Epper 1983), allowing for a clearer identification of the primordia of each adult structure. At the beginning of this period, 44 h APF, we already observe the 3 main tissue groups described in McQueen et al. (2022): the analia, epigynium/hypogynium, and oviprovectum/vagina (Fig. 1a).

The 2 halves of the adult female analia exhibit distinct morphologies. At early timepoints, the analia is donut-shaped and divided horizontally into the epiproct on the dorsal side and the hypoproct on the ventral side (Fig. 1a' and e'). As development progresses, the epiproct extends and assumes a beak-like shape, while the hypoproct becomes blunted (Fig. 1a' to d'). The analia is joined to the genitalia by the epigynium, a tissue that encircles the epiproct and hypoproct (Fig. 1e and f'). At 52 h APF, the epigynium starts to protrude laterally (Fig. 1c' and d') and will eventually cover the anterior end of the hypogynium.

The most prominent tissue of the adult female genitalia is the hypogynium, which consists of 2 flat plates that form the lateral sides of the genitalia. During pupal development, the initially rounded hypogynium straightens while increasing in length and area (Green et al. 2019). The plates of the hypogynium are connected dorsally, ventrally, and interiorly by the membranous oviprovectum (Fig. 1e and f'). At 44 h APF, the oviprovectum is a smooth tissue, and over

time, both the dorsal and ventral hypogynial membranes condense and pull the hypogynial plates closer together by 56 h APF (Supplementary Fig. 1a to d). The vagina, which represents the anterior internal portion of the genitalia, appears cylindrical at 44 h APF (Fig. 1a). Over developmental time, 5 vaginal folds are established: 1 dorsal (the vaginal furcal dorsal fold), a dorsolateral pair (the vaginal furcal dorsolateral folds), and a lateral pair (the vaginal furcal lateral folds) (Fig. 1c and d, Supplementary Fig. 1c' and d'). These folds change the vaginal morphology from a hollow structure to one that has little vacant medial space.

Validating the hybridization chain reaction in male genital tissues

We previously measured the expression patterns of 100 transcription factors in the male pupal genitalia at 48 h APF using colorimetric in situ hybridization (Vincent et al. 2019). Here, we used hybridization chain reaction (HCR) to measure expression patterns in the female and male pupal genitalia for 29 transcription factors that exhibited strong spatial patterning in male samples (Dirks and Pierce 2004; Duckhorn et al. 2022). Among other advantages, HCR enables the collection of 3D images using confocal microscopy. We first validated the efficacy of the protocol by comparing our HCR signal in male genitalia to the previous signal by colorimetric in situ hybridization (Vincent et al. 2019). At 48 h APF, signals for 28 out of 29 genes showed similar expression patterns for both methods (Supplementary Table 1 and Fig. 1). The only notable difference was observed for *Polycomblike* (*Pcl*), which had ubiquitous expression in the case of HCR, while our previous results indicated that *Pcl* expression was restricted to the phallus.

Overall, our HCR signal exhibited reduced background and an increased range of detection compared to previous results and revealed features of genital expression patterns that were not previously noted. For example, while previous analyses of the *Drop* (*Dr*) expression pattern were restricted to the analia, medial clasper (ie surstylis), and the hypandrium (Vincent et al. 2019), HCR revealed additional *Dr* expression in the lateral plate (epandrial ventral lobe), ventral clasper, and posterior lobe (epandrial posterior lobe) (Supplementary Fig. 1 and Table 1). The concordance between our HCR results and the previous in situ hybridization analysis suggested that we could successfully deploy HCR to measure 3D gene expression patterns in female pupal genitalia.

Transcription factors with ubiquitous expression patterns

Eight transcription factors displayed ubiquitous or nearly ubiquitous expression patterns in the genitalia of both males and females. In one case, the gene *hairy*, gene expression in the males was ubiquitous, but was notably absent from the epigynium in female samples (Supplementary Fig. 2).

In most cases of ubiquitous expression, although present in all tissues, gene expression was stronger in some tissues than others. Observation of tissue-level variation in the ubiquitous expression is indicated in Table 1 and Supplementary Table 1. One illustrative example is the gene *bowl*, which shows low levels of expression across the entire genitalia in both males and females, but is strongly expressed in the dorsal parts of the female genitalia and the external structures of the male genitalia (Supplementary Figs. 1 and 2).

Expression patterns in the female and male analia

The analia comprise the dorsal-most tissues of the terminalia in both sexes. The majority of the transcription factors we screened showed patterned expression in the analia of one or both sexes (Fig. 2, Supplementary Fig. 3, Table 1, Supplementary Table 1).

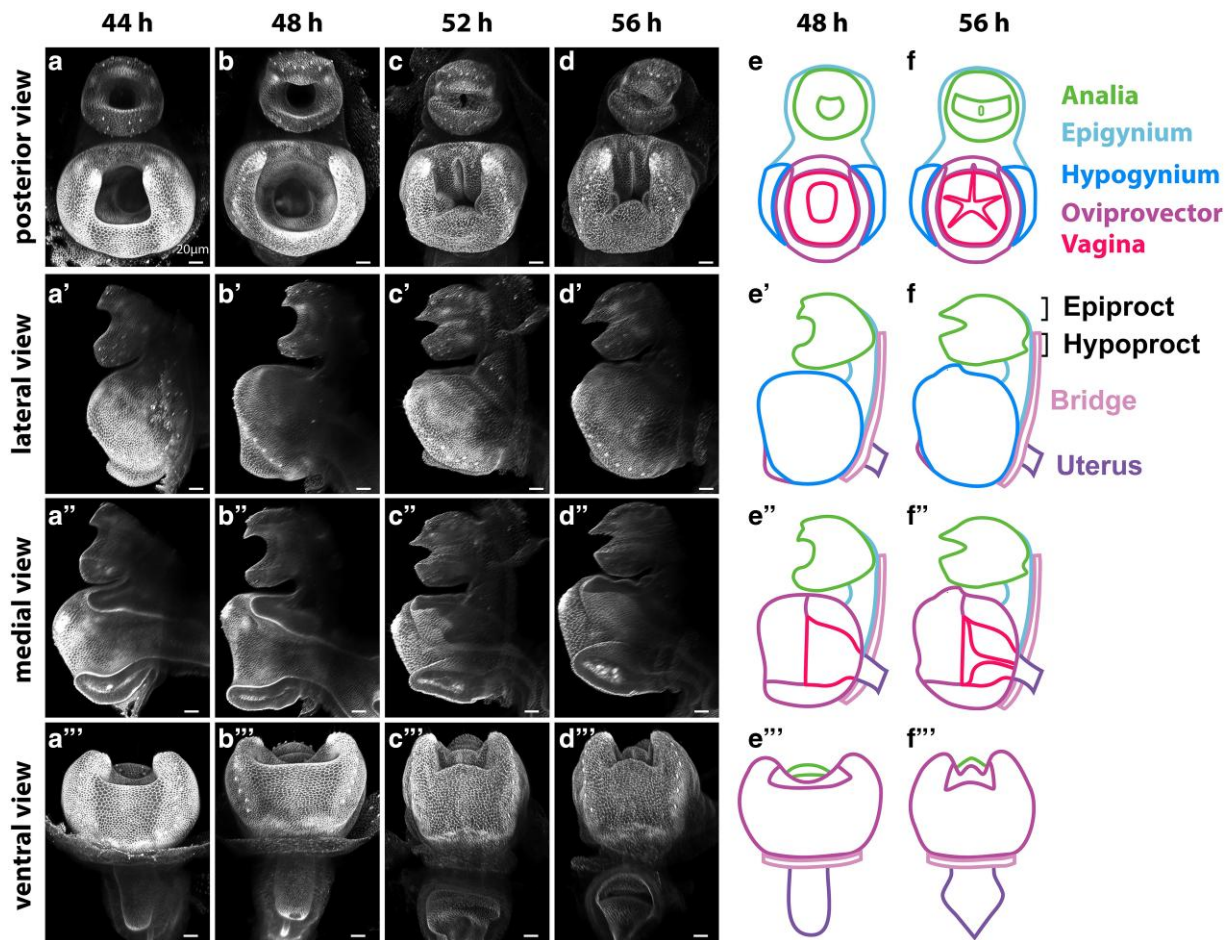


Fig. 1. Time course of female terminalia development. *armadillo*-GFP fluorescence marks the apical cellular junctions and outlines the overall shape of developing female terminalia. a, a', a'', a''') Posterior, lateral, medial, and ventral views of the terminalia at 44 h APF. At this time point, the analia is tubular with a depression between the dorsal analia (epiproct) and ventral analia (hypoproct). The genitalia are also tubular with a depression on the dorsal side where the dorsal oviprovector membrane is found. b, b', b'', b''') At 48 h APF, the epiproct becomes more pointed at its posterior edge. The ventral oviprovector membrane begins to separate from the hypogynium, and a small pocket starts to form in the dorsolateral area of the vagina. c, c', c'', c''') At 52 h APF, the epiproct becomes more pointed, and its distance relative to the hypoproct decreases. The oviprovector membrane begins to form folds that create distinct pockets along its surface. The oviprovector ventral membrane also condenses medially and pulls the hypogynium closer together. d, d', d'', d''') At 56 h APF, the epiproct and hypoproct close further, and the folds of the oviprovector membrane increase in depth. e, e', e'', e''', f, f', f'', f''') Schematic representations of 48 h APF and 56 h APF samples illustrate the analia (green), epigynium (light blue), hypogynium (dark blue), oviprovector (purple), and vagina (red). Black brackets designate the position of the epiproct and hypoproct. Scale bars represent 20 μ m for all images.

The gene *cad* (Fig. 2c) was previously identified as a marker of the anal plate in males (Moreno and Morata 1999; Vincent et al. 2019). Likewise, *cad* exhibits strong expression throughout the female analia (Fig. 2c). Although the analia of both sexes are derived from the 10th abdominal segment, in females, the anal plate is divided horizontally, which produces 2 distinct substructures. In males, the anal plate is divided vertically and exhibits bilateral symmetry (Green et al. 2019) (Fig. 2b). Despite this difference, some of the genes we examined are patterned similarly in both sexes. For example, *Dr* is expressed in the anus in both males and females (Table 1, Supplementary Table 1, Fig. 2c, Supplementary Fig. 1), as is *hairy*, *AbdB*, *cad*, and *noc* (Supplementary Figs. 1 and 3). *HLHm3* exhibits medial expression in both the female epiproct and hypoproct (Fig. 2c), which mimics its similar expression pattern in the male anal plate, except that the pattern is rotated 90° (Supplementary Fig. 1). Moreover, in both males and females, the genes *drm*, *sob*, and *kni* are expressed in a third, distinct tissue located between the 2 lobes of the anal plate in males and between the epiproct and hypoproct in females. We assign the term “paraproct” to this structure (Fig. 2c).

Some genes, including *E5*, *ems*, and *inv*, are expressed distinctly in either the epiproct or hypoproct in females and also exhibit dorsal/ventral patterning in males (Supplementary Figs. 1, 3, and 6). Conversely, some genes are markers of specific subsections of the female analia and have distinct patterns when compared to the males. For instance, *C15* is restricted to the epiproct (Ridgway, Figueras Jimenez, et al. 2024), while the *fd96Cb* marks the hypoproct (Fig. 2c). We did not detect *C15* in the analia of males, whereas *fd96Cb* is present in males but only in the ventral paraproct (Supplementary Fig. 1).

Expression patterns in the female external genitalia: hypogynium and epigynium

The female analia is connected to the genitalia via a cuticular structure called the epigynium (Fig. 3a). Aside from ubiquitously expressed genes, we found that few of the screened transcription factors exhibited patterned expression in the epigynium (Table 1). The transcription factors *en* and *inv* (Fig. 3b) are expressed strongly throughout the epigynium; these patterns wrap around the anterior side of the analia to the hypogynial plates (Supplementary

Table 1. Summary of transcription factor expression in female genital structures.

Gene	Regions of expression in the female genitalia											
	Analia			External genitalia			Oviprovector and internal reproductive structures					
	Ubiquitous	Epiproct	Paraproct	Hypoproct	Anus	Epigynium	Hypogynium	Oviprovector Dorsal	Oviprovector Lateral/Ventral	Hypogynial anteroventral bridge	Vagina	Uterus
<i>Abd-B</i>		0	0	0	1	2	2	2	2	2	1	1
<i>bowl</i>	x	1	1	1	1	1	2	2	2	...	1	...
<i>C15</i>		2	0	0	0	0	0	0	0	...	0	...
<i>cad</i>		2	2	2	1	0	0	0	0	...	0	0
<i>crol</i>	x	2	1	2	1	0	2	1	1	...	1	...
<i>drm</i>		2	2	0	0	0	0	0	0	2	0	...
<i>Drop</i>		0	0	0	2	0	2	0	0	...	0	0
<i>dsx</i>		2	2	2	0	0	2	2	2	...	2	...
<i>E5</i>		1	2	2	0	0	2	0	2	...	0	2
<i>ems</i>		1	1	1	0	0	2	0	1	...	0	2
<i>en</i>		1	0	0	0	2	0	2	2	...	2	...
<i>esg</i>		1	0	1	0	0	0	0	2	...	2	...
<i>fd96Cb</i>		0	2	2	0	0	0	0	0	...	0	...
<i>grh</i>	x	2	1	2	0	1	2	2	1	...	1	...
<i>gsb</i>		0	0	0	0	0	0	0	2	...	0	...
<i>hairy</i>		1	1	1	2	0	2	2	1	...	1	1
<i>HLHm3</i>		2	2	2	0	0	2	0	2	...	2	...
<i>inv</i>		1	0	0	0	2	0	2	2	...	2	...
<i>kni</i>		2	2	2	0	1	1	2	2	...	1	...
<i>lola</i>	x	2	2	2	1	1	2	2	2	...	1	...
<i>maf-S</i>	x	1	1	1	0	0	1	1	1	...	1	...
<i>mod</i>	x	2	1	1	1	0	1	1	2	...	2	...
<i>noc</i>		1	2	1	2	0	1	0	2	...	0	...
<i>opa</i>		0	0	1	0	0	2	0	0	...	0	...
<i>Pcl</i>	x	2	2	2	1	0	2	2	2	...	2	...
<i>Poxn</i>		0	0	0	0	0	0	0	2	...	0	...
<i>retn</i>		0	1	0	0	0	0	1	1	...	2	...
<i>sob</i>		1	2	1	0	1	1	1	1	2	0	2
<i>ttk</i>	x	1	2	1	1	0	0	2	2	...	1	...

Image stacks of 48 h APF HCR signal were manually inspected, and expression levels were qualitatively assigned to 1 of 4 categories for each structure in the developing female genitalia: strongly expressed (2); weakly expressed (1); not expressed (0); and could not be determined from available data (–). Genes exhibiting ubiquitous expression are also noted, with additional information regarding tissue-level variation in expression indicated in the table as needed.

Fig. 4). Notably, this pattern in which *en* and *inv* expression wraps around the anal plate is also observed in males (Supplementary Fig. 1). The gene *kni* is also weakly expressed in parts of the epigynium (Supplementary Fig. 2).

The hypogynium plays a key functional role during oviposition in many pest species and is one of the best-studied female genital structures in *Drosophila* (Atallah et al. 2014). For instance, previous work traced *Drosophila suzukii*'s increased hypogynial length—a derived state that facilitates more penetrant oviposition in that species—to altered cellular morphogenesis (Green et al. 2019). Understanding the developmental landscape of this structure is therefore of substantial interest. We found that many transcription factors in our panel exhibit patterned expression in the hypogynium. For example, *opa* exhibits strong expression along the entire posterior edge of each hypogynial plate, while *kni* is restricted to the anterior region (Fig. 3c). The genes *E5* and *ems* are also expressed throughout the anterior and ventral parts of the hypogynium, with some weak expression along the ventral posterior edge (Fig. 3c). *Drop* expression is restricted to a small dorso-lateral patch on each plate (Fig. 3c). Interestingly, co-staining of *Dr* and *E5* reveals that the *Dr* expression pattern is anticorrelated with *E5*, except for a very small section of overlap between the 2 patterns (see Fig. 5c). A similar phenomenon occurs between *E5* and *Dr* in males, where the 2 genes are expressed in separate regions except for a small region of overlap in the epandrial ventral lobe (see Fig. 5c). The expression pattern for *HLHm3* in the hypogynium is restricted to a stripe of expression along the posterior edge

of each plate (Fig. 3c). This pattern begins at the dorsal posterior tip of the plates and then bifurcates partway along the posterior edge, with 1 stripe of expression continuing along the posterior edge and the other continuing in a ventral trajectory slightly further toward the anterior portion of the tissue. In adults, the cuticle in this area of each hypogynial plate is partly, but not entirely, bisected—a feature called the hypogynial mid-dorsal incision (McQueen et al. 2022). The expression pattern of *HLHm3* may be associated with the patterning of the mid-dorsal incision, which could be investigated in samples at later time points.

Expression patterns in the oviprovector and vagina

The oviprovector is a membranous tissue interior to the hypogynium that connects the hypogynium to the vagina (Supplementary Fig. 4a and d). Although contiguous at early stages of development, the adult oviprovector comprises 2 parts—a dorsal membrane and a ventral membrane. The ventral membrane also extends dorsally and harbors a set of scales (McQueen et al. 2022). In *Drosophila erecta* and *D. teissieri*, the dorsal oviprovector has evolved pocket and shield-like structures that may protect against copulatory harm (Kamimura and Mitumoto 2012; Kamimura 2016).

Many of the transcription factors we surveyed are expressed in the developing oviprovector. For example, *en* and *inv* are expressed throughout the dorsal oviprovector and partially extend ventrally (Fig. 4d), whereas the *Poxn* expression pattern is restricted to the medial portion of the dorsal oviprovector (Fig. 4d).

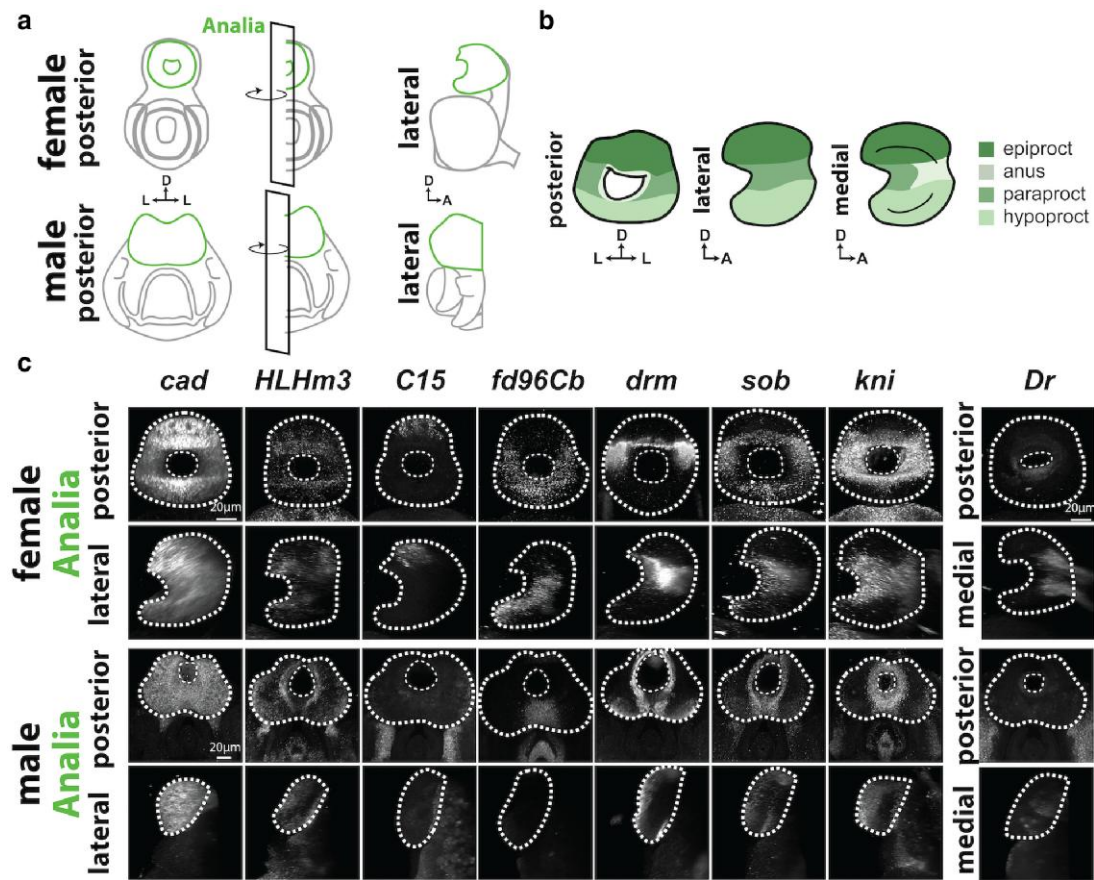


Fig. 2. Transcription factors expressed in the analia. a) Schematics of the *D. melanogaster* female and male terminalia with the analia highlighted in green. For lateral views, the left half of the terminalia is masked, and the sample is turned to face the lateral side. b) Schematic of the tissues within the female terminalia: epiproct (dark green), anus (gray-green), paraproct (sage green), and hypoproct (light green). c) HCR signal in female and male 48 h APF analia, with detected signal rendered in white. Dashed lines outline the analia (thick outline) and the anus (thin outline). *cad* is expressed throughout the epiproct and hypoproct. Dashed lines outline the analia (outer circle) and the anus (inner circle). *HLHm3* is expressed in both the epiproct and hypoproct. *C15* is restricted to the epiproct, while *fd96Cb* is restricted to the paraproct and hypoproct. *drm*, *sob*, and *kni* are mainly expressed in the paraproct. *Dr* is expressed in the anus, which connects the analia to the rectum.

gsb exhibits a highly specific expression pattern at a location that may represent the boundary between the dorsal and lateral/ventral regions of the ovipositor (Fig. 4d). *noc* is strongly expressed in the dorsolateral portion of the ovipositor, while *E5* and *ems* are expressed in the ventral ovipositor (Fig. 4d).

The vagina is located anterior (with respect to the whole animal) to the developing ovipositor. Although it is difficult to determine the exact boundary between the developing ovipositor and vagina at 48 h APF, some transcription factors are clearly expressed in the anterior region associated with vaginal tissue. A key example is *retn* (Fig. 4e). In contrast, *esg* is expressed in regions of the anterior (more internal) ovipositor as well as the most posterior portions of the vagina (Fig. 4e) in a pattern that suggests a possible association with some of the eventual vaginal furcal folds. In these internal tissues, *HLHm3* is expressed in 2 lateral stripes that extend anteriorly from the ovipositor into the vagina (Fig. 4e). This expression pattern may mark a boundary between tissue types that differentiate later in development.

We found that 2 transcription factors, *drm* and *sob*, serve as markers for the hypogynial anteroventral bridge, a tissue that connects the terminalia to the abdominal tergite 7 (Fig. 4b). Although this named structure in adults is restricted to a ventral fragment (McQueen et al. 2022), our results suggest that the hypogynial anteroventral bridge shares developmental origins with the anterior edge of the epigynum, which wraps around the anterior

portion of the analia. For many transcription factors that we screened, we were unable to determine with confidence whether they were expressed in the ventral portion of the hypogynial anteroventral bridge (see Table 1).

In this study, the uterus was only retained in a subset of dissected samples, which allowed us to determine whether some transcription factors were expressed in this tissue. For example, *sob* (Fig. 4c), *ems* (Supplementary Fig. 5), and *E5* were all strongly expressed in the uterus.

Co-expression of transcription factors in males and females

HCR accommodates multiplexing—i.e. detecting RNA for multiple genes in the same sample. Multiplexing allows researchers to define precise expression domains of genes relative to each other with high confidence. For instance, multiplexed *AbdB* and *cad* HCR signals reveal that these 2 transcription factors are directly adjacent and mutually exclusive (Fig. 5a). Previous expression data suggested that *cad* expression was limited to the anal plate, and *AbdB* was absent in that tissue (Vincent et al. 2019). We find that *AbdB* also has low-level expression in the anus and the inner tissue of the analia of females and males, but this pattern remains adjacent to and distinct from *cad*-expressing cells (Fig. 5a). In addition to this mutually exclusive gene pair, *kni* expression in the inner ring of the analia is adjacent to a *mod* expression domain in the

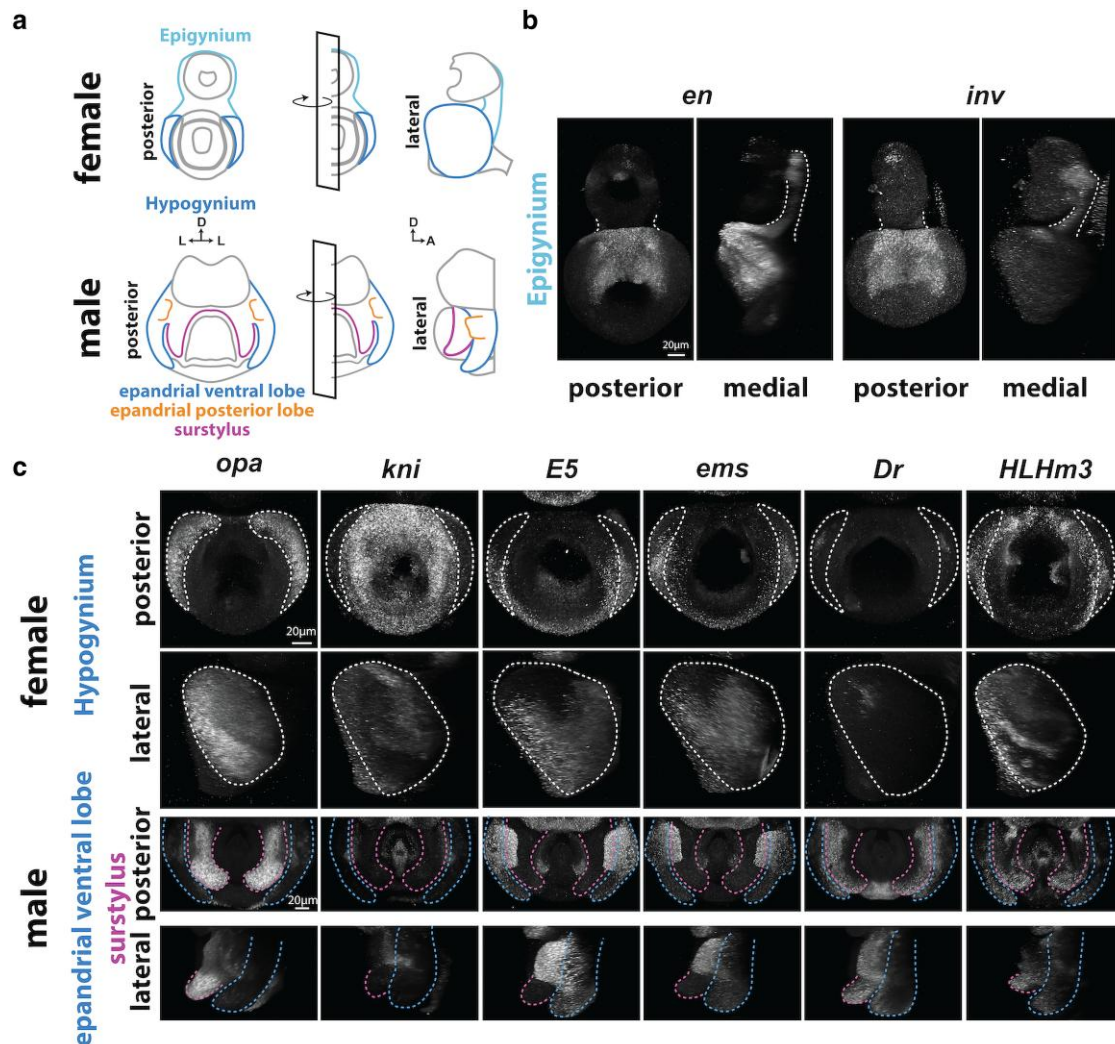


Fig. 3. Transcription factors expressed in the epigynum and hypogynum. a) Schematics of the female and male terminalia with the female epigynum (light blue), and hypogynum (dark blue), as well as the male epandrial ventral lobe (dark blue), epandrial posterior lobe (orange), and surstylus (purple) highlighted. b) HCR signal in the female epigynum, where the detected signal appears white. *en* and *inv* are expressed across the hypogynum. c) HCR signal in the female 48 h APF hypogynum (dashed line) and male epandrial ventral lobe (thick dashed line) and surstylus (thin dashed lines). Posterior (top) and lateral views (bottom) are shown. *opa* is expressed in the posterior edge of the hypogynum, while *kni* expression is restricted to the anterior region of the hypogynum. *ems* and *E5* are expressed broadly in the epandrial ventral lobe but decrease in expression midway through the anterior-posterior axis. *Dr* is expressed in the posteroventral hypogynum. *HLHm3* is expressed throughout the dorsal hypogynum and in a punctate pattern in the ventral hypogynum. For males, the epandrial posterior lobe blocks portions of the surstylus in the lateral view.

epiproct (Fig. 5b), while *Dr* is expressed directly dorsal to *E5* in the hypogynal posterior lobe (Fig. 5c). Finally, in the vagina, *HLHm3* is expressed between 2 adjacent *esg* expression domains (Fig. 5d). We also find examples of co-expressed transcription factors, as *Poxn*'s medial expression is entirely within the broader *en* expression in the ovipositor dorsal membrane (Fig. 5e). These genes are also co-expressed in males, specifically in tissues surrounding the phallus and the ventral portion of the clasper (Supplementary Fig. 1). These examples of mutual exclusion and co-expression may indicate negative and positive regulatory interactions respectively, which can be interrogated further in future experimental work.

Discussion

In this work, we present a detailed 3D atlas of transcription factors expressed in developing genital structures in *D. melanogaster*. We collected 3D images of genital morphology at 4 time points during

pupal development in females, and we measured the expression patterns of 29 transcription factors at 48 h APF in both females and males. This analysis confirmed and extended previous results found using colorimetric in situ hybridization in males (Vincent et al. 2019) and revealed a complex gene regulatory landscape in female tissues. Our results suggest that some morphological observations in a recent detailed analysis of adult female genitalia may have molecular underpinnings (McQueen et al. 2022). For example, we found that certain transcription factors can serve as marker genes in the analia (*cad*), hypogynum (*opa*), and vagina (*retn*). We also found that expression patterns in the female analia are subdivided along the dorsal/ventral axis, compared with the medial/lateral organization of the male analia. Finally, multiplexing of hybridization chain reactions allowed us to label expression domains of transcription factor pairs in single tissues, which suggested putative regulatory interactions between them. These results suggest that the same transcription factors may have functional roles in genital development in both sexes, and further

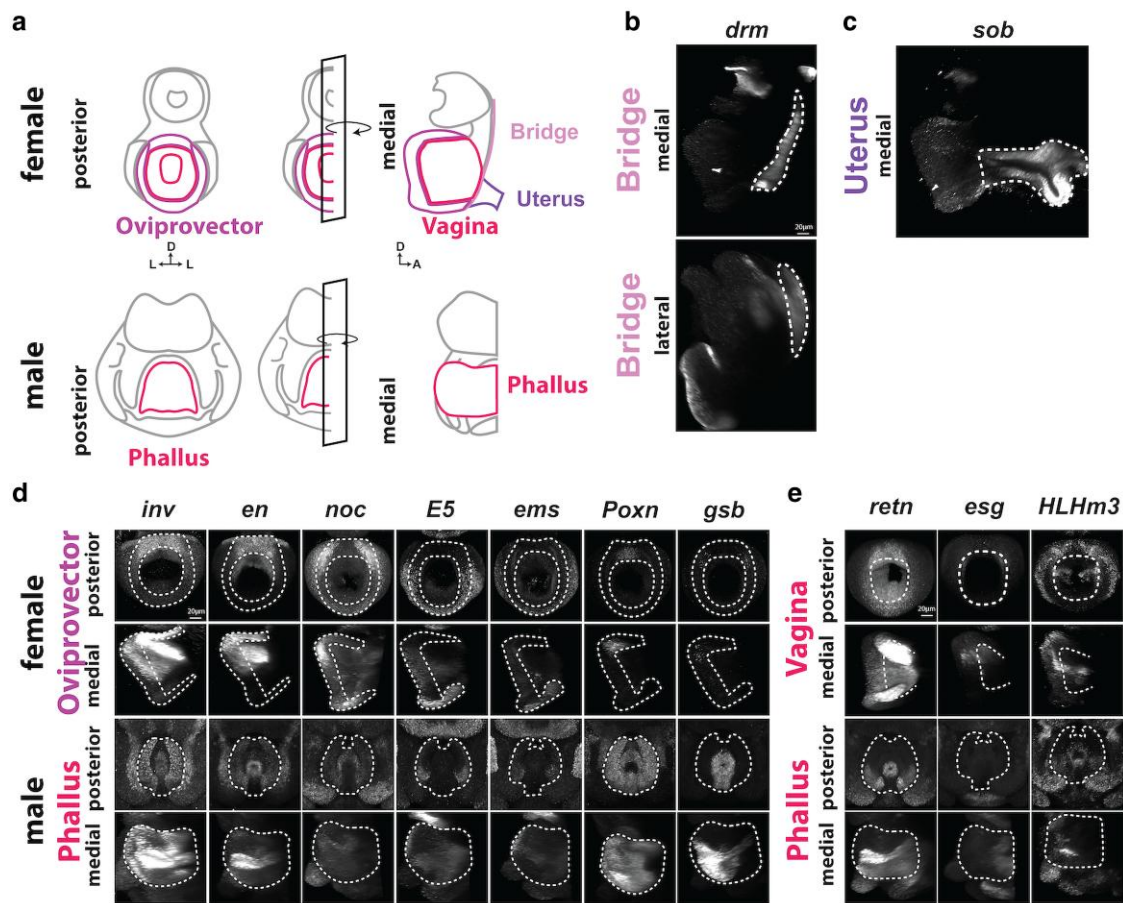


Fig. 4. Transcription factors expressed in the ovipositor and internal genitalia. a) Schematics of the female and male terminalia with the female ovipositor (light purple), vagina (red), bridge (pink), and uterus (dark purple) highlighted in females and the male phallus highlighted in red. b) HCR signal in the bridge (dashed line) for both sexes. c) HCR signal in the female uterus (dashed line). c) HCR signal in the female 48 h APF ovipositor (dashed line) and male phallus (dashed line), where the detected signal appears in white. *inv*, *en*, and *noc* are expressed in the dorsal ovipositor, while *inv* and *en* are also expressed in the dorsal vagina. *E5* and *ems* are expressed in the lateral and ventral regions of the ovipositor. *Poxn* is expressed in the dorsomedial ovipositor, and *gsb* is expressed in the dorsolateral ovipositor. d) HCR signal in the female vagina (dashed line) and male phallus. *retn* is expressed throughout the vagina. *esg* and *HLHm3* are expressed in the dorsal lateral portion of the vagina.

investigation into their regulation may provide genetic reagents for future work on morphogenesis, sexual dimorphism, and co-evolution.

Drosophila appendages have long been powerful model systems to study how gene regulatory networks operate during development. Recent work has illuminated the cellular mechanisms underlying the formation and evolution of genital structures in females and males: Changes in ovipositor morphology have been traced to cellular expansion and rearrangements in females (Green et al. 2019), while the male posterior lobe forms as a result of extreme cell elongation mediated by the apical extracellular matrix (Smith et al. 2020). Transcription factors like *Pox neuro* and *Sox21b* have been functionally characterized in male genital structures (Glassford et al. 2015; Ridgway, Hood, et al. 2024), but their role in females is comparatively understudied. As we measured the 3D expression patterns for 29 transcription factors in female samples, future work may investigate how those genes regulate their target effector proteins to coordinate the complex cellular processes operating during genital development.

Of note, our gene expression signals are currently restricted to a single time point during the pupal stage in both sexes, and these expression patterns likely shift as development progresses (Vincent et al. 2019). The functional role of these transcription factors on genital development in both sexes can be assessed using

genetic perturbations (Ridgway, Figueras Jimenez, et al. 2024; Tanaka et al. 2022), although these experiments may be hindered by the limited number of drivers that have been characterized in these tissues (Robinett et al. 2010; Chatterjee et al. 2011; Green et al. 2019; Ridgway, Figueras Jimenez, et al. 2024; Ridgway, Hood, et al. 2024). Many of these transcription factors are also expressed in other appendages, but many studies have focused on only 1 tissue or timepoint in development (Supplementary Tables 3 and 4). A comprehensive analysis of the expression patterns of all developing appendages during both larval and pupal development would give us better expectations for the extent of genetic network overlap between serially homologous organs. Genetic experiments may also confirm putative regulatory interactions that we have inferred by observing mutually exclusive or overlapping gene expression patterns. Finally, a complete picture of the gene regulatory landscape in female genital tissues is possible through single-cell transcriptomics, and this current atlas may prove useful for interpreting such data by providing marker genes for annotating cell populations after tissue dissociation.

This study shows that many transcription factors that have been found to pattern the male terminalia also pattern the female terminalia, even though the adult structures take on radically different morphologies. Future work will be required to determine how the same regulatory genes produce these divergent body

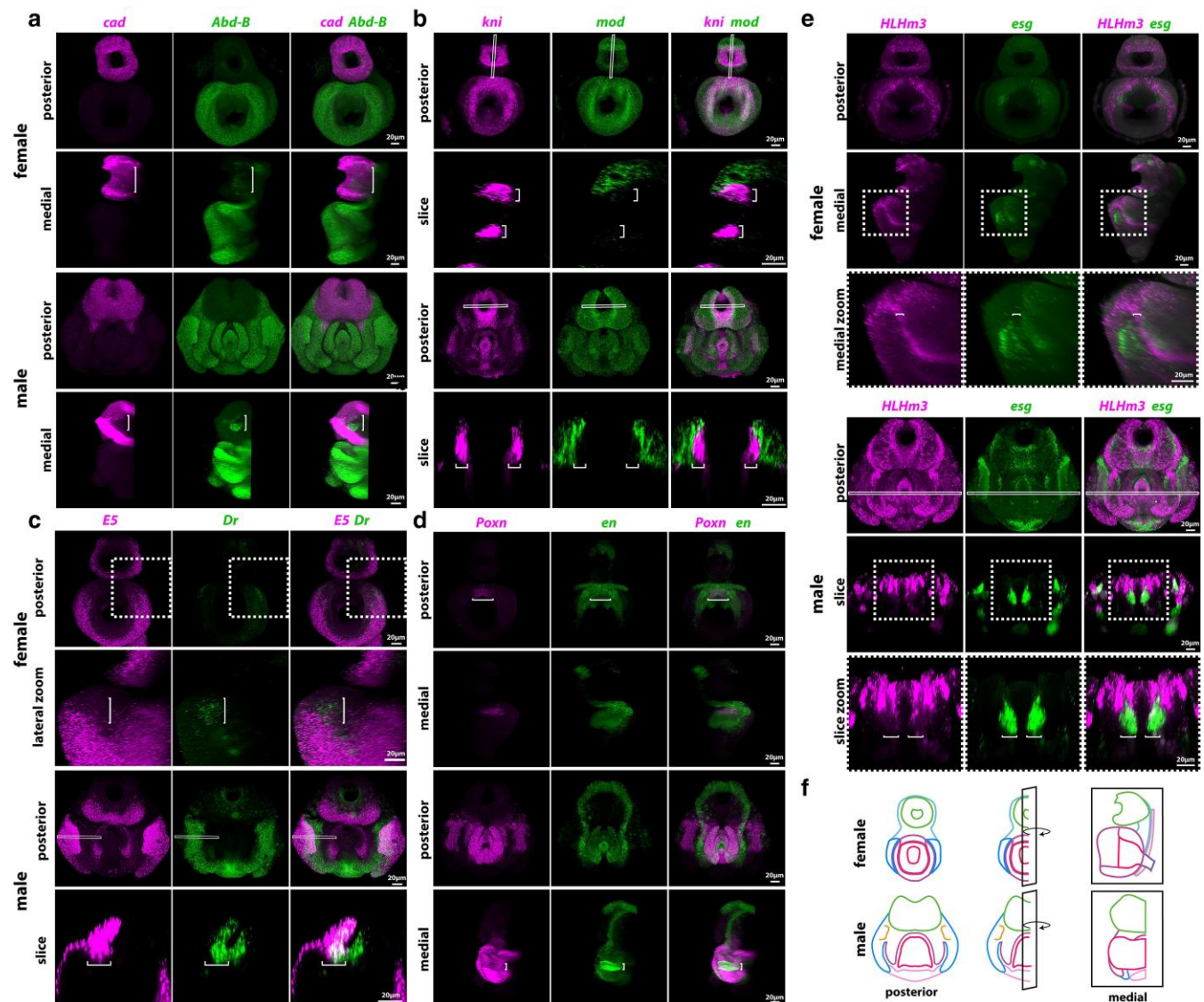


Fig. 5. Multiplexed HCR reveals fine-scale relationships between gene expression patterns. a) HCR signal for *cad* (magenta) and *Abd-B* (green) in female and male 48 h APF pupal genitalia. In females, *cad* is expressed in the hypoproct and epiproct but not in the anus (white brackets). *Abd-B* is expressed in the genitalia as well as the anus (brackets) in both sexes. b) HCR signal for *kni* (magenta) and *mod* (green). *kni* is expressed in the female inner epiproct (brackets) but not in the outer portion (brackets), while *mod* is expressed in the outer epiproct and not the internal portion. The male analia shows a similar pattern to the females, but rotated 90°. c) HCR signal for *E5* (magenta) and *Dr* (green). *Dr* is expressed in the hypogynal posterior lobe, while *E5* is expressed ventrally relative to the *Dr* expression domain (brackets). In the male epandrial posterior lobe, both *E5* and *Dr* are co-expressed along the medial edge, while only *E5* is expressed on the lateral edge. d) HCR signal for *Poxn* (magenta) and *en* (green). *Poxn* expression is restricted in the medial portion of the female oviprovect dorsal membrane (brackets), while *en* is expressed throughout the oviprovect dorsal membrane. *Poxn* is expressed throughout the male phallus but is restricted from the tube that runs through it (known as the phallotreme) while *en* shows strong expression in the phallotreme (brackets). e) HCR signal for *HLHm3* (magenta) and *esg* (green). *HLHm3* and *esg* are both expressed in the female oviprovect ventral membrane in close proximity. A close-up of the indicated region (dashed box) reveals that these expression patterns do not overlap. *HLHm3* expression splits 2 *esg* expression domains (brackets). The male phallus shows anterior expression of *esg* (brackets) directly adjacent to posterior *HLHm3* expression pattern. f) Schematic of the posterior and medial views of the *D. melanogaster* terminalia. For female terminalia, we highlight the analia (green), epigynium (light blue), hypogynium (dark blue), oviprovect (light purple), vagina (red), bridge (pink), and uterus (dark purple). For male terminalia, we highlight the analia (green), epandrial ventral lobe (dark blue), epandrial posterior lobe (orange), surstylus (purple), phallus (red), and hypandrium (pink). Solid boxes indicate the location of an optical slice through the samples.

parts. The imaginal discs that produce most appendages of the fly share serial homology. Homeotic transformations can cause both large and small shifts in morphology. Legs can be transformed into antennae (Percival-Smith et al. 1997), and forelegs can be transformed into midlegs through the same homeotic process (Lewis et al. 1980). In contrast to all other fly imaginal discs, the genital disc is composed of primordia from 3 different abdominal segments, each of which forms separate compartments. Abdominal segment 10 produces the analia of both males and

females, while abdominal segment 9 will produce the genitalia in males and is reduced to only form the accessory glands in females (Keisman et al. 2001). Abdominal segment 8 produces the genitalia in females and is reduced to only the epandrial anterodorsal phragma in males (Keisman et al. 2001). Comparing directly homologous abdominal segments between the sexes is therefore unlikely to be fruitful due to these extreme reductions. Determining top-tier patterning genes that establish the unique genetic cascades in the male and female genitalia will help us

better understand if their level of serial homology is high, like forelegs and midlegs, or low, like legs and antennae.

One of our most striking findings is the parallel and perpendicular expression patterns of the male and female analia, respectively. In this work, we found that the expression patterns of *HLHm3*, *drm*, and *kni* appear similar in both sexes (Supplementary Fig. 6) but rotated by 90° and oriented orthogonally when female samples are compared to male samples. Our observations raise additional questions concerning the developmental origins of sexual dimorphism in the analia: when and how are the axes of symmetry specified in both sexes? Because both female and male genital structures evaginate during metamorphosis in the pupal stage, observing physical markers of symmetry prior to evagination is technically challenging, though previous work suggests that divisions in the anal plate are detectable in females by 12 h APF (Epper 1983). Future work should examine the expression patterns of differentially patterned transcription factors in early pupal samples to determine if they exhibit bilateral symmetry prior to evagination.

Conversely, we find transcription factors that pattern the dorsal (*en* and *inv*) and ventral (*E5* and *ems*) regions of the analia of both sexes (Supplementary Fig. 6). These differences in gene expression may be connected to differences between the sexes in how the anal plate is physically divided into substructures. These results indicate that these 2 structures may be homologous even if they are found in perpendicular planes. Therefore, we suggest that the region between the anal plates in males also be referred to as the paraproct. There are also transcription factors that pattern the paraproct in only 1 sex: *retn* is expressed in females, while *esg* is expressed in males (Supplementary Fig. 6). These differences in expression may indicate sex-specific differences in tissue organization or morphology. The stark change in the orientation of the analia makes the paraproct an exciting model of sexual dimorphism. The males of some species, such as *Drosophila ananassae*, show a division between the dorsal and ventral regions of their cercus, which raises the question of whether the horizontal or vertical orientation is ancestral (Bock and Wheeler 1972; Polak and Rashed 2010).

As the knowledge gap between female and male terminalia closes, we will gain a more nuanced view of how these 2 tissues are connected to one another developmentally. A recent study found that the transcription factors *C15* (also analyzed in this study), *Al*, and *lim* were all expressed in both female and male terminalia (Ridgway, Figueras Jimenez, et al. 2024). A male-specific enhancer for *C15* was identified, indicating that the regulation of the male and female terminalia expression patterns is independent (Ridgway, Figueras Jimenez, et al. 2024). The separation of female and male regulatory sequences would decrease pleiotropy, allowing for the expression pattern of each sex to evolve independently. Similarly, an analysis of single-nucleus RNA-seq of the adult female reproductive tract revealed that 40% of the previously characterized male-specific seminal fluid genes are expressed in the female reproductive tract (Thayer et al. 2024). The lack of attention to female traits has led us to mislabel genes as male-specific and provided us with an overly simplistic view of sexual dimorphism. We must continue to build resources in female-specific systems if we hope to understand how sexually dimorphic traits evolve.

Data availability

Supplemental material comprised of images of additional samples in posterior, medial, and lateral views is available through Figshare via DOI: [10.25386/genetics.30307171](https://doi.org/10.25386/genetics.30307171), and the confocal stacks of the HCR data are available via the University of

Pittsburgh Institutional Repository and can be accessed via the DOI: <https://doi.org/10.18117/vbwn-4116>. Additional files are available upon request.

Supplemental material available at [GENETICS](https://www.genetics.org) online.

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